

Sakhawat Hossan

Ph.D. Student in Computer Science | University of North Carolina at Greensboro
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RESEARCH INTERESTS

Reliability, security, and efficiency of large language models, hallucination detection and multimodal AI, AI-mediated communication detection, autonomous multi-agent systems, efficient inference and edge computing. My research combines machine learning with over nine years of software development experience to build trustworthy and resource-efficient AI systems.

EDUCATION

Ph.D. in Computer Science — in progress	2023–Present
University of North Carolina at Greensboro, Greensboro, NC	
Advisor: Dr. Jing Deng. Research focus: Reliability, security, and efficiency of large language models.	
M.S. in Computer Science	05/2026
University of North Carolina at Greensboro, Greensboro, NC	
Master in Information Technology	12/2016
Institute of Information Technology, University of Dhaka, Bangladesh	
B.S. in Computer Science and Engineering	12/2013
Dhaka International University, Dhaka, Bangladesh	

TEACHING EXPERIENCE

Course Instructor	Spring 2026
Department of Computer Science, University of North Carolina at Greensboro	
CSC 250-02 — Foundations of Computer Science I	
- Independently designed the course materials and taught the entire semester-long, in-person course, with full responsibility for course planning, instruction, assessment, and grading. - Developed instructional materials and organized content across 13 modules, aligning lessons and assessments with course learning objectives. - Delivered three 50-minute class meetings each week covering proof techniques and induction, sets, functions and relations, graphs and trees, recursion, algorithm complexity, counting, and recurrence relations. - Designed and administered five assignments, two exams, and a cumulative final examination; evaluated student work and managed course grades. - Managed course materials and announcements through Canvas, established course policies, and provided individual support through office hours and appointments.	

TEACHING INTERESTS

Discrete mathematics and foundations of computer science, artificial intelligence and machine learning, natural language processing, introductory programming in Python, software engineering and web development, database systems.

PUBLICATIONS

Journal Articles

- (J1) Tavila Sharmin, **Sakhawat Hossan**, and Rohan A. Shirwaiker. "Design and Validation of a Multi-Modal Bioreactor System: Assessing the Effects of Perfusion and Cyclic Tensile Stimulation on Mechanical and Biological Properties of 3D-Printed Missing-Rib Auxetic Scaffolds," Bioengineering, vol. 13, no. 2, Art. no. 140, 2026.
doi: [10.3390/bioengineering13020140](https://doi.org/10.3390/bioengineering13020140)
- (J2) **Sakhawat Hossan** and Naushin Nower. "Fog-based dynamic traffic light control system for improving public transport," Public Transport, vol. 12, pp. 431–454, 2020.
doi: [10.1007/s12469-020-00235-z](https://doi.org/10.1007/s12469-020-00235-z)

Conference Papers

- (C1) Qian Tong, **Sakhawat Hossan**, and Jing Deng. "A Modality-Agnostic Definition of Multi-Modal Hallucination," 2026 IEEE International Conference on Artificial Intelligence, Internet of Things and Cloud Computing (AlloT), pp. 448–453, 2026.
Doi: [10.1109/AlloT68874.2026.11569576](https://doi.org/10.1109/AlloT68874.2026.11569576)
- (C2) Naveena Pokala, **Sakhawat Hossan**, Jing Deng, and Dayong Huang. "Stock Prediction based on Annual Reports using Large Language Model," 2026 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA), 2026.
doi: [10.1109/ACDSA67686.2026.11467931](https://doi.org/10.1109/ACDSA67686.2026.11467931)
- (C3) **Sakhawat Hossan** and Jing Deng. "Detection of LLM Hallucinations Using Late Internal Representations," 2025 International Conference on Machine Learning and Applications (ICMLA), pp. 1403–1410, 2025.
doi: [10.1109/ICMLA66185.2025.00214](https://doi.org/10.1109/ICMLA66185.2025.00214)
- (C4) **Sakhawat Hossan** and Jing Deng. "MMEC: Using MoE Efficiently for Mobile Edge Computing," 2025 16th International Conference on Information and Communication Technology Convergence (ICTC), pp. 1631–1636, 2025.
doi: [10.1109/ICTC66702.2025.11387890](https://doi.org/10.1109/ICTC66702.2025.11387890)
- (C5) **Sakhawat Hossan**, Jing Deng, Rebecca Schwendler, and Charles P. Egeland. "Identifying Social Interaction Outliers with the Use of Network Analysis: Disk Decoration in the Middle Magdalenian Period," 17th International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2025), Lecture Notes in Computer Science, vol. 16323, pp. 28–35, 2025.
doi: [10.1007/978-3-032-13821-7_3](https://doi.org/10.1007/978-3-032-13821-7_3)
- (C6) Mahmuda Mahfuz Rimi, Md. Rasel Ahmed, **Sakhawat Hossan**, Md. Masudur Rahaman, Md. Anowar Hossen Faysal, and Md. Ferdouse Ahmed Foysal. "A Cost-Effective IoT Device for Protecting People from the Dangers of Environmental Risk in Bangladesh," 2022 8th International Conference on Computing Engineering and Design (ICCED), 2022.
doi: [10.1109/ICCED56140.2022.10010393](https://doi.org/10.1109/ICCED56140.2022.10010393)
- (C7) Naushin Nower and **Sakhawat Hossan**. "Efficient Dynamic Traffic Light Control for ITS," Proceedings of the Second International Conference on Internet of Things, Data and Cloud Computing (ICC '17), Article 190, pp. 1–7, 2017.
doi: [10.1145/3018896.3056807](https://doi.org/10.1145/3018896.3056807)

PROFESSIONAL SERVICE

Program Committee Member and Reviewer	2026
IEEE International Conference on Tools with Artificial Intelligence (ICTAI)	
Reviewer	2026
ICCC — artificial intelligence, machine learning, and related computing area	
IEEE Open Journal of the Computer Society (OJCS)	

RESEARCH EXPERIENCE

Research Assistant

08/2023–Present

Department of Computer Science, UNC Greensboro | Advisor: Dr. Jing Deng

- Develop hallucination detection methods using late internal representations, output-level signals, and multimodal consistency analysis.
- Investigate AI-mediated communication detection through text-generation signals, author consistency, interaction behavior, and active challenges.
- Explore Mixture-of-Experts architectures, quantization, and resource-aware inference; study autonomous agent behavior, collaboration, and emerging security risks.
- Construct datasets and evaluate transformer-based models using PyTorch, Hugging Face Transformers, and GPU computing.

Research Assistantship

2016–2020

Institute of Information Technology, University of Dhaka, Bangladesh

Earlier research: Developed sensor-driven traffic-control algorithms for individual and multiple intersections, using wireless sensor networks, fog computing, and a Docker-based distributed architecture.

INDUSTRIAL EXPERIENCE

Senior Software Programmer

07/2017–08/2023

Fiber@Home Ltd, Dhaka, Bangladesh

- Developed internal ERP systems as a full stack developer using Python, PHP, JavaScript, C#, and web frameworks; developed Android mobile applications.

Web Developer

04/2013–03/2016

Server Oasis Ltd

- Developed themes and dynamic websites with DotNetNuke, WordPress, and Drupal, using PHP, JavaScript, C#, Bootstrap, and AngularJS.

TECHNICAL SKILLS

Programming and data: Python, PHP, JavaScript, C#, SQL/MySQL, HTML/CSS, Pandas, and NumPy.

AI and research computing: PyTorch, Hugging Face Transformers, GPU/CUDA computing, text classification, representation learning, authorship analysis, vision-language models, quantization, and model evaluation.

Software systems: Full stack development, ERP integration, debugging, compatibility testing, Android development, Docker, and distributed fog computing.

HONORS AND AWARDS

2013 | Dean's Scholarship, Dhaka International University. GPA of 4.00/4.00 in three consecutive semesters.

2011 | First Runner-Up, Victory Day Programming Contest, Dhaka International University.

2009 | Winner, National Science Fair, district level.

LEADERSHIP AND MEMBERSHIPS

Assistant General Secretary and Treasurer, Dhaka International University Alumni Association.

Member: IEEE; Bangladesh Computer Society (BCS); The American Center, Dhaka, Bangladesh.

REFERENCES

Dr. Jing Deng — UNC Greensboro | jing.deng@uncg.edu

Dr. Minjeong Kim — UNC Greensboro | mkim@uncg.edu